

Kinematics: Reference Frames & Acceleration Composition

1 Reference Frames & Coordinate Setup

We consider two distinct coordinate reference frames:

- **Absolute Frame (R):** Fixed global reference frame centered at $B(O, x, y, z)$.
- **Relative Frame (R'):** Moving/rotating frame centered at $O'(x', y', z')$.

2 Motion Equations (Linear + Rotational)

The motion combines linear expansion along a local axis with uniform angular rotation around the z -axis:

- **Linear Relative Motion:** Motion along the local x' -axis at constant relative speed V_r :

$$\vec{r}' = \begin{pmatrix} V_r \cdot t \\ 0 \\ 0 \end{pmatrix}$$

- **Rotational Motion:** Uniform rotation around the z -axis with constant angular velocity ω :

$$\theta(t) = \omega \cdot t \implies t = \frac{\theta}{\omega}$$

- **Absolute Position Trajectory:** Combining radial movement and rotation yields the ground frame position vector:

$$\vec{r}(t) = \begin{pmatrix} x(t) \\ y(t) \\ z(t) \end{pmatrix} = \begin{pmatrix} V_r \cdot t \cdot \cos(\omega t) \\ V_r \cdot t \cdot \sin(\omega t) \\ 0 \end{pmatrix}$$

3 Law of Composition of Accelerations

The absolute acceleration vector $\vec{a}_a$ is decomposed into three components:

$$\vec{a}_a = \vec{a}_r + \vec{a}_e + \vec{a}_c$$

3.1 Relative Acceleration ($\vec{a}'_r$)

The acceleration evaluated strictly inside the moving frame R' :

$$\vec{a}'_r = \vec{0} \quad \left(\text{since } V_r = \text{constant} \implies \frac{d\vec{V}_r}{dt} = 0 \right)$$

3.2 Entrainment / Centripetal Acceleration ($\vec{a}'_e$)

The acceleration component resulting from the frame's rotation:

$$\vec{a}'_e = \vec{\Omega} \wedge (\vec{\Omega} \wedge \vec{r}') = \begin{pmatrix} -\omega^2 \cdot V_r \cdot t \\ 0 \\ 0 \end{pmatrix}$$

3.3 Coriolis Acceleration ($\vec{a}'_c$)

The dynamic interaction term between relative velocity and frame rotation:

$$\vec{a}'_c = 2\vec{\Omega} \wedge \vec{V}_r = \begin{pmatrix} 0 \\ 2\omega \cdot V_r \\ 0 \end{pmatrix}$$

4 Total Acceleration in Moving Frame (R')

Summing the Coriolis and entrainment components inside R' yields:

$$\vec{a}'_a = \vec{a}'_c + \vec{a}'_e = \begin{pmatrix} -\omega^2 \cdot V_r \cdot t \\ 2\omega \cdot V_r \\ 0 \end{pmatrix}$$

5 Frame Transformation ($R' \rightarrow R$)

To express the acceleration vector in the fixed absolute frame R , we apply the standard 2D rotation matrix $R_z(\omega t)$:

$$R_z = \begin{pmatrix} \cos(\omega t) & -\sin(\omega t) & 0 \\ \sin(\omega t) & \cos(\omega t) & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Final Absolute Acceleration Vector ($\vec{a}_a$): Computing $\vec{a}_a = R_z \times \vec{a}'_a$ gives the complete absolute acceleration vector relative to the ground observer:

$$\vec{a}_a = \begin{pmatrix} -2\omega \cdot V_r \cdot \sin(\omega t) - \omega^2 \cdot V_r \cdot t \cdot \cos(\omega t) \\ 2\omega \cdot V_r \cdot \cos(\omega t) - \omega^2 \cdot V_r \cdot t \cdot \sin(\omega t) \\ 0 \end{pmatrix}$$